Hengyu Lian

Chapel Hill, North Carolina

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Research Interests

- **Computational optics:** including imaging, photography, display (holography), and optical neural network

Education

- **University of North Carolina at Chapel Hill** — **Chapel Hill, NC**
  Doctor of Philosophy — *August 2024 – May 2029 (expected)*
  Major: Computer Science
  Advisor: ***Prof. Praneeth Chakravarthula***
- **Shanghai Jiao Tong University** — **Shanghai, China**
  Master of Science and Engineering — *September 2021 – June 2024*
  Major: Electronic and Information Engineering
  Advisor: ***Prof. Yuan Qu and Prof. Jiamiao Yang***
- **Tianjin University** — **Tianjin, China**
  Bachelor of Engineering — *September 2017 – July 2021*
  Major: Measurement and Control Technology and Instrument
  Advisor: ***Prof. Min Lai***

Publications

- **Hengyu Lian**, Debabrata Mandal, Anzhou Wen, et al., "Conditional Imaging for Privacy-Preserving Visual Sensing", arXiv preprint arXiv:2610.28835, 2026.
- Yuan Qu†, **Hengyu Lian†**, Chunxu Ding, et al., "High frame-rate reconfigurable diffractive neural network based on superpixels", Optics Letters, 2023, 48(19):5025-5028.
- Yuan Qu†, **Hengyu Lian†**, Rongjun Shao, et al., "Time series analysis for financial indices using optical reservoir computing", Optical Engineering, 2024, 63(5): 054108-054108.
- Zihao Zou, Jiaming Liu, **Hengyu Lian**, et al., "Asynchronous Multimodal View Synthesis with Probabilistic Temporal Encoding", SIGGRAPH Asia 2026 Conference Papers, 2026.
- Huaizhi Qu†, Yujie Wang†, Ruichen Zhang, **Hengyu Lian**, et al., "HoloZip: High Hologram Compression via Latent-of-Latent Coding." In 2025 IEEE International Conference on Computational Photography (ICCP), pp. 1-10. IEEE, 2025.
- Huazhen Liu, Rongjun Shao, Yuan Qu, Chunxu Ding, **Hengyu Lian**, et al. "High-accuracy phase unwrapping based on binarized wrap count", Optics Express, 2024, 32, 44605-44623.

Skills and Awards

- **Skills:** Python, MATLAB, PyTorch, SolidWorks, Optical Experimental Skills
- **Awards:**
  - Guo Xie Birong Scholarship rewarded by Shanghai Jiao Tong University, 10/2023 (Rank: 1/62)
  - The 1st Scholarship rewarded by Shanghai Jiao Tong University, 11/2023, 11/2022 (top30%)

Work Experience and Services

- **Reviewer:** *Present*
  Optics Letters, Optics Express
- **Graduate Research Assistant:** *August 2024 – Present*
  Visual Computing and Augmented Intelligence Lab
  Advisor: Prof. Praneeth Chakravarthula
- **Teaching Assistant:** *August 2025 – Present*
  Course: Americans with Disabilities Act (ADA)
  Advisor: Prof. Brent Munsell
- **Teaching Assistant:** *August 2024 – May 2025*
  Course: COMP 590-059 Programming Methods, Models, Languages and Analysis
  Advisor: Prof. David Stotts
- **Graduate Research Assistant:** *September 2021 – June 2024*
  Intelligent Photoelectric Sensing Institute
  Advisor: Prof. Yuan Qu and Prof. Jiamiao Yang